



Practical - 27: Name of the main topic 1

107) WAP to demonstrate the use of private, public, protected and default access modifiers using two package. (A)

```
package p1;

public class Demo {

    public int a = 10;
    protected int b = 20;
    int c = 30;          // default
    private int d = 40;

    public void display() {
        System.out.println("Public = " + a);
        System.out.println("Protected = " + b);
        System.out.println("Default = " + c);
        System.out.println("Private = " + d);
    }
}
```

Package p2

```
package p2;

import p1.Demo;

public class Test {

    public static void main(String[] args) {

        Demo obj = new Demo();

        System.out.println("Public = " + obj.a);

        // Not accessible
        // obj.b;
        // obj.c;
        // obj.d;
    }
}
```

108) Assume that there are two packages, student and exam. A student package contains Student class and the exam package contains Result class. Write a program that generates mark sheet for students. (B)

```
package student;

public class Student {
```



```
public int rollNo;
public String name;

public Student(int rollNo, String name) {
    this.rollNo = rollNo;
    this.name = name;
}
}
```

Exam Package:

```
package exam;

import student.Student;

public class Result extends Student {

    int m1, m2, m3;

    public Result(int rollNo, String name,
        int m1, int m2, int m3) {

        super(rollNo, name);

        this.m1 = m1;
        this.m2 = m2;
        this.m3 = m3;
    }

    void display() {

        int total = m1 + m2 + m3;
        double per = total / 3.0;

        System.out.println("Roll No : " + rollNo);
        System.out.println("Name : " + name);
        System.out.println("Total : " + total);
        System.out.println("Percentage : " + per);
    }

    public static void main(String[] args) {

        Result r = new Result(
            101,
            "Aniruddh",
            80,
            90,
            85);

        r.display();
    }
}
```

109)	Define a class A in package apack. In class A, three variables are defined of access modifiers protected, private and public. Define class B in package bpack which extends A and write display method which accesses variables of class A. Define class C in package cpack which has one method display() in that create one object of class A and display its variables. Define class ProtectedDemo in package dpack in which write main () method. Create objects of class B and C and class display method for both these objects. (C)
	<pre>package apack; public class A { protected int x = 10; private int y = 20; public int z = 30; }</pre>
	<pre>package bpack; import apack.A; public class B extends A { public void display() { System.out.println("Protected = " + x); System.out.println("Public = " + z); // y not accessible } }</pre>
	<pre>package cpack; import apack.A; public class C { public void display() { A obj = new A(); System.out.println("Public = " + obj.z); // obj.x not accessible // obj.y not accessible } }</pre>



```
package dpack;

import bpack.B;
import cpack.C;

public class ProtectedDemo {

    public static void main(String[] args) {

        B b = new B();
        b.display();

        C c = new C();
        c.display();

    }
}
```